

Multivariate Change-Point Detection Using Feature-Based Inductive Conformal Martingales

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Abstract

Change-point detection in sequential data can be formulated as the problem of identifying violations of exchangeability as observations arrive over time. In this work, we study feature-based change-point detection within the framework of Inductive Conformal Martingales, which are well known for providing theoretical validity guarantees on the probability of false alarms. We construct nonconformity measures that quantify how unusual the features of a test instance are. In particular, our proposed nonconformity measure is based on an ellipsoidal k -nearest neighbour density estimator combined with Mahalanobis distance.

We evaluate the proposed method on four real-world datasets in which an artificial change-point is inserted, as well as on an air-quality dataset where the goal is to estimate possible sensor recalibration points. We also compare our approach with two competing nonparametric multivariate change-detection methods with false-alarm control mechanisms. The results illustrate the potential of feature-based conformal martingales for practical sequential change-point detection in multivariate data.

Keywords: change-point detection, conformal martingales, exchangeability, nonconformity measures, feature extraction

1. Introduction

Change-point detection in sequential data is a fundamental problem in statistics and machine learning, with applications in domains such as healthcare, finance, and environmental monitoring (Aminikhanghahi and Cook, 2017). The goal is to detect, as quickly as possible, deviations from a reference distribution while controlling the rate of false alarms.

Formally, a change-point can be defined as a time instant at which the data-generating distribution changes. Consider a stream $S = \{x_0, x_1, \dots\}$, where each x_i is an observation. If S can be partitioned into $S_{0,t}$ and $S_{t+1,\dots}$, each generated by a different distribution, then a change-point has occurred at time $t + 1$. In this study, we focus on the unlabeled setting, where labels are not available.

One principled framework for sequential change detection is provided by Inductive Conformal Martingales (Volkhonskiy et al., 2017). These methods detect violations of the exchangeability assumption in the data stream and raise an alarm when the martingale

exceeds a threshold corresponding to a pre-specified significance level. Since exchangeability is expected to hold under a stable distribution, a detected violation provides evidence of a change-point. Conformal martingales are particularly attractive because they rely on minimal assumptions about the data and provide theoretical validity guarantees on false alarms.

In this work, we propose a feature-based approach for change-point detection using Inductive Conformal Martingales. Specifically, we design a nonconformity measure based on an ellipsoidal k -nearest-neighbour density estimator that combines local density information with Mahalanobis distance. This nonconformity measure aims to capture local geometric and density variations in the data, leading to improved sensitivity to distributional changes in multivariate streams.

We evaluate the proposed method on several real-world datasets, where change-points are artificially introduced, as well as on an air-quality dataset where the objective is to identify potential sensor recalibration points. We compare the proposed conformal martingale detector with QuantTree (Boracchi et al., 2018) and QT-EWMA (Frittoli et al., 2023), two nonparametric multivariate change-detection methods with false-alarm control mechanisms. QuantTree controls the false positive rate in a batch/window setting, QT-EWMA controls the average run length under the null, whereas conformal martingales provide anytime-valid false-alarm control through Ville’s inequality (Ville, 1939).

Contributions. The main contributions of this paper are:

- We introduce a novel feature-based nonconformity measure for Inductive Conformal Martingales based on an ellipsoidal k NN density estimator and Mahalanobis distance.
- We provide an empirical evaluation on multiple real-world datasets, including a practical application in air-quality monitoring.
- We demonstrate that the proposed approach achieves good detection performance and a low false alarm rate while retaining the theoretical validity of conformal martingales.

The remainder of this paper is organized as follows. Section 2 reviews related work on change-point detection and conformal methods. Section 3 presents the conformal martingale framework. Section 4 introduces the proposed feature-based nonconformity measure. Section 5 describes the experimental setting and reports the results. Section 6 concludes the paper.

2. Related Work

This section reviews related work on change-point detection, with emphasis on online multivariate methods, nonparametric approaches, and conformal martingales.

2.1. Change-Point Detection

Change-point detection has been studied in statistics, machine learning, and data mining. The main objective is to identify time points at which the data-generating distribution changes. Classical approaches include parametric methods such as likelihood-ratio tests and cumulative sum procedures, as well as non-parametric and machine-learning-based

methods. Surveys such as (Aminikhanghahi and Cook, 2017) provide broad overviews of supervised and unsupervised change-point detection methods for time-series data.

These methods can be divided into two main categories: offline and online change-point detection. In the offline setting, the whole sequence is analyzed after it has been observed. In contrast, in the online setting, observations are processed sequentially, and the aim is to detect a change as soon as possible after it occurs. In this work, we focus on the online setting, where observations arrive one at a time and the detector must operate without access to future data.

2.2. Nonparametric Multivariate Change Detection

Detecting changes in multivariate streams is more challenging than in univariate settings, especially when the pre-change distribution is unknown. Parametric methods can be powerful when their assumptions hold, but they may fail when the data distribution is complex or unknown. Nonparametric methods are therefore attractive because they make fewer assumptions about the data-generating distribution. For this reason, we focus in particular on QuantTree, a nonparametric multivariate change-detection method that provides probabilistic control of the false positive rate and is therefore close in spirit to the validity guarantees of conformal martingales (Vovk et al., 2005). We also discuss QT-EWMA, an online extension of QuantTree that provides guarantees in terms of ARL_0 , the expected time before the first false alarm.

QuantTree was introduced as a histogram-based method for detecting changes in multivariate data streams (Boracchi et al., 2018). Its main idea is to construct histogram bins using binary splits, so that test statistics defined over the resulting histogram have a distribution that does not depend on the unknown stationary distribution. This allows detection thresholds to be computed in a distribution-free way and enables control of the false positive rate (FPR).

In the batch setting, QuantTree provides a distribution-free procedure for setting the detection threshold so that the FPR is bounded by a pre-specified level α . Thus, α plays the role of a significance level for testing whether a window of observations is generated from the reference distribution.

However, QuantTree is originally designed for batch or window-based testing rather than sequential monitoring. Therefore, it does not directly provide online false-alarm control over an indefinitely long stream. In addition, its performance depends on the choice of the number of bins and the window size. If the number of bins is too small, the histogram may fail to capture distributional changes, whereas too many bins may lead to unreliable estimates when the training set is limited. Finally, as a histogram-based method, QuantTree may lose sensitivity in high-dimensional settings unless dimensionality reduction or a feature representation that preserves change-relevant information is used.

QT-EWMA (Frittoli et al., 2023) extends QuantTree to the online setting by combining QuantTree histograms with an exponentially weighted moving average statistic. Its main guarantee is control of the average run length under the null hypothesis, denoted by ARL_0 , which is the expected number of observations before the first false alarm. Thus, QT-EWMA provides probabilistic false-alarm control for online multivariate monitoring, in contrast to

the original QuantTree method, which controls the false positive rate in a batch/window setting.

These methods are closely related to our work because they address nonparametric multivariate change detection, although their false-alarm criteria differ from the anytime-valid guarantee provided by conformal martingales.

2.3. Conformal Martingales for Change Detection

Conformal martingales provide a principled framework for online testing of exchangeability. The idea of testing exchangeability online using martingales was introduced by [Vovk et al. \(2005\)](#). Under exchangeability, conformal p-values are uniformly distributed, and a betting function can be used to construct a nonnegative martingale. Large martingale values provide evidence against exchangeability and can therefore be interpreted as evidence of a change in the data-generating distribution.

Martingale-based change detection has been applied to both labeled and unlabeled data streams. In the labeled case, nonconformity measures are often derived from classifiers or nearest-neighbour rules. Since this work focuses on unlabeled data, we next review studies that extend martingale-based change detection to the unsupervised setting.

In the unlabeled setting, [Ho and Wechsler \(2007\)](#) extended the martingale methodology using clustering-based nonconformity measures and showed its feasibility for detecting changes in unlabeled data streams. They also proposed a multiple-martingale test using different feature views, showing that the sensitivity of the martingale method can be improved by using multiple representations of the data.

In addition, [Volkhonskiy et al. \(2017\)](#) introduced an Inductive version of Conformal Martingales for change-point detection. In their approach, an initial set of observations is used as a training set, and subsequent nonconformity scores are computed with respect to this fixed training set. They experimented with several betting functions and nonconformity measures, including nearest-neighbour and likelihood-ratio scores. Their results were competitive with classical methods such as CUSUM, Shiryaev-Roberts, and posterior probability statistics. While this work demonstrated the effectiveness of ICMs for change-point detection, its empirical evaluation was mainly based on synthetic 1D-Gaussian settings.

In a previous work ([Eliades and Papadopoulos, 2024b](#)), we applied Inductive Conformal Martingales to change-point detection on aCGH genomic data. The method required only preprocessing and did not assume a specific model or distribution for the data. Two ICM-based algorithms were proposed: one using preprocessed log2-ratio values as nonconformity scores, and another incorporating both label-conditional and label-distribution information, where labels represented genomic gains and losses. The results showed that incorporating this additional information improved change-point detection compared with the simpler ICM approach and produced competitive results against CBS and ED. While that work focused on one-dimensional genomic sequences after preprocessing, the present paper studies feature-based nonconformity measures for multivariate change-point detection using an ellipsoidal k NN density estimator combined with Mahalanobis distance.

3. Inductive Conformal Martingales

This section presents the basic concepts of Inductive Conformal Martingales (ICMs) and explains how nonconformity scores, conformal p-values, and exchangeability martingales are constructed in the unlabeled change-point detection setting considered in this work.

3.1. Exchangeability

Let $(Z_1, Z_2, \dots)$ be an infinite sequence of random variables. The joint distribution $\mathbb{P}(Z_1, \dots, Z_N)$ is said to be exchangeable if it is invariant under any permutation of its variables. An infinite sequence $(Z_1, Z_2, \dots)$ is exchangeable if, for every $N \in \mathbb{N}$, the marginal distribution $(Z_1, \dots, Z_N)$ is exchangeable.

Exchangeability is closely related to the i.i.d. assumption through de Finetti's theorem (Schervish, 1995), which states that every exchangeable distribution can be represented as a mixture of i.i.d. distributions. Thus, testing exchangeability can be viewed as testing whether the data-generating mechanism remains stable.

3.2. Exchangeability Martingales

A test exchangeability martingale, hereafter referred to simply as an exchangeability martingale, is a sequence of nonnegative random variables $(S_1, S_2, S_3, \dots)$ satisfying $\mathbb{E}(S_{n+1} | S_1, \dots, S_n) = S_n$ under the null hypothesis of exchangeability.

Ville's inequality states that for any threshold $C > 0$,

$$\mathbb{P}(\exists n : S_n \geq C) \leq \frac{1}{C}.$$

Thus, large martingale values provide evidence against exchangeability. For example, $S_n \geq 100$ corresponds to rejecting exchangeability at the 1% significance level.

3.3. Calculating Nonconformity Scores and P-values

Let $\{x_1, x_2, \dots\}$ be a sequence of observations, where each x_i is represented by a feature vector. ICM uses the first k observations, $\{x_1, \dots, x_k\}$, as a training set for constructing nonconformity scores for subsequent observations.

For each new observation $x_j \in \{x_{k+1}, x_{k+2}, \dots\}$ we compute a nonconformity score $\alpha_j = A(x_j, \{x_1, \dots, x_k\})$ where A is a nonconformity measure (NCM). Larger scores indicate that an observation is less conforming to the rest of the set. Thus, distributional changes are expected to alter the distribution of the nonconformity scores and consequently distort the conformal p-value distribution away from uniformity.

For each new example x_j , define $H_j = \{\alpha_{k+1}, \alpha_{k+2}, \dots, \alpha_j\}$.

The corresponding conformal p-value is

$$p_j = \frac{|\{\alpha_i \in H_j : \alpha_i > \alpha_j\}| + U_j |\{\alpha_i \in H_j : \alpha_i = \alpha_j\}|}{j - k}, \quad (1)$$

where $U_j \sim U(0, 1)$ is used to break ties and ensure validity (Vovk et al., 2003).

Under exchangeability, the p-values are i.i.d. Uniform(0, 1).

3.4. Constructing Exchangeability Martingales

Given a sequence of p-values $(p_1, p_2, \dots)$, the martingale is defined as

$$S_n = \prod_{i=1}^n f_i(p_i), \quad (2)$$

where f_i is a betting function satisfying

$$\int_0^1 f_i(p) dp = 1, \quad f_i(p) \geq 0.$$

Under exchangeability, this implies

$$\mathbb{E}(S_{n+1} \mid S_0, S_1, \dots, S_n) = S_n.$$

Proposition 1 *If $\int_0^1 f_i(p) dp = 1$ then under any exchangeable distribution it holds:*

$$\mathbb{E}(S_{n+1} \mid S_0, S_1, \dots, S_n) = S_n$$

Proof [Proof of Proposition 1] The integral $\int_0^1 f_i(p) dp$ equals 1

We will now prove that the conditional expectation is preserved under any exchangeable distribution:

$$\begin{aligned} \mathbb{E}(S_{n+1} \mid S_0, S_1, \dots, S_n) &= \int_0^1 \prod_{i=1}^n f_i(p_i) \cdot f_{n+1}(p) dp \\ &= \prod_{i=1}^n f_i(p_i) \cdot \int_0^1 f_{n+1}(p) dp \\ &= \prod_{i=1}^n f_i(p_i) = S_n \end{aligned} \quad (3)$$

■

Thus (S_n) is a test martingale.

The betting function can be a density estimator of the empirical p-value distribution. As shown by [Fedorova et al. \(2012\)](#), if the estimator is consistent, plug-in martingales asymptotically achieve at least the growth rate of any fixed betting function.

Using (2), the martingale can be updated recursively as

$$S_n = S_{n-1} f_n(p_n).$$

If $S_n = M > 1$, then by Ville's inequality exchangeability can be rejected at significance level $1/M$.

To avoid numerical underflow, we compute the martingale in logarithmic scale.

4. Proposed Approach

This section presents the proposed feature-based change-point detection methodology. We first describe how change-points are detected using Inductive Conformal Martingales. We then describe the betting function used to update the martingale. Finally, we introduce the proposed nonconformity measure based on k -nearest-neighbour density estimation and Mahalanobis distance.

4.1. Change-Point Detection using ICM

To detect a change-point at a pre-specified significance level δ , the martingale value must exceed the threshold $1/\delta$, which leads to rejection of the exchangeability assumption. By Ville’s inequality, this controls the probability of false alarms.

Specifically, if the martingale value S_i exceeds 100, a change-point is detected at significance level 1%. The detection procedure is summarized in Algorithm 1. Here, L denotes the number of previous p-values used by the betting function estimator.

Algorithm 1: Change-point detection using ICM

Input: Training set $\{x_1, \dots, x_k\}$, monitoring stream $\{x_{k+1}, \dots, x_n\}$, significance level δ

Output: Change-point alarms

Initialize $S_0 \leftarrow 1$

for $i = 1$ **to** $n - k$ **do**

 Compute nonconformity score

$\alpha_i \leftarrow A(x_{k+i}, \{x_1, \dots, x_k\})$

 Compute conformal p-value

$p_i = \frac{|\{j: \alpha_j > \alpha_i\}| + U_i |\{j: \alpha_j = \alpha_i\}|}{i}$

 Estimate betting function

$h_i = h(p_{i-L}, \dots, p_{i-1})$

 Update martingale

$S_i \leftarrow S_{i-1} h_i(p_i)$

if $S_i > 1/\delta$ **then**

 | Raise change-point alarm

end

end

4.2. Betting Function

In this work, we use the cautious betting function introduced in (Eliades and Papadopoulos, 2024a), which avoids betting when there is insufficient evidence against exchangeability. This prevents the martingale from collapsing to values close to zero and can improve sensitivity to future changes.

Let f_n^j , $j = 1, \dots, M$, denote a set of candidate betting functions satisfying

$$\int_0^1 f_n^j(p) dp = 1, \quad f_n^j(p) \geq 0.$$

For each candidate j , define the corresponding martingale process

$$S_n^j = \prod_{i=1}^n f_i^j(p_i).$$

At time n , the cautious betting function h_n is defined as

$$h_n(p) = \begin{cases} 1, & \text{if } \max_{j \in \{1, \dots, M\}} \left(\frac{S_{n-1}^j}{\min_{1 \leq k \leq W} S_{n-k}^j} \right) \leq \epsilon, \\ f_n^m(p), & \text{otherwise,} \end{cases} \quad (4)$$

where

$$m = \arg \max_{j \in \{1, \dots, M\}} \left(\frac{S_{n-1}^j}{\min_{1 \leq k \leq W} S_{n-k}^j} \right).$$

Here, W is the look-back window, $\epsilon > 0$ is a threshold, and M is the number of candidate betting functions. The ratio compares the current value of each candidate martingale with its recent minimum over the last W steps. Thus, the method bets only when at least one candidate has shown sufficient recent growth. Otherwise, it sets $h_n(p) = 1$, meaning that the martingale remains unchanged at that step.

In the experiments, we use the same candidate betting functions as in (Eliades and Papadopoulos, 2024a). Specifically, the cautious betting strategy is combined with three interpolated histogram density estimators with 5, 10, and 15 bins, and three nearest neighbour density estimators with 5, 10, and 15 neighbours. These estimators are fitted to the empirical distribution of the previously observed conformal p-values.

4.3. Feature-Based Nonconformity Measure

The main contribution of this work is a feature-based nonconformity measure designed for multivariate change-point detection. The proposed NCM combines local density estimation through k -nearest neighbours with Mahalanobis distance that captures the local geometry of the data.

Given a training set $\{x_1, \dots, x_n\} \subset \mathbb{R}^d$ and a new observation x , we first identify its k -nearest neighbours in the reference set, denoted by

$$N_K(x) = \{x_{(1)}, \dots, x_{(K)}\}.$$

Using these neighbours, we estimate a local covariance matrix

$$S_x = \text{Cov}(N_K(x)) + \lambda I_d, \quad (5)$$

where $\lambda > 0$ is a regularization parameter ensuring invertibility.

To capture the local shape of the neighbourhood, distances are measured using the Mahalanobis metric induced by S_x :

$$d_M(x_i, x) = \sqrt{(x_i - x)^T S_x^{-1} (x_i - x)}. \quad (6)$$

We define the local ellipsoidal radius as the maximum Mahalanobis distance among the K neighbours,

$$t(x) = \max_{x_i \in N_K(x)} d_M(x_i, x). \quad (7)$$

This induces an ellipsoid centered at x with volume

$$V(x) = V_d t(x)^d \sqrt{\det(S_x)}, \quad (8)$$

where $V_d = \frac{\pi^{d/2}}{\Gamma(d/2+1)}$ is the volume of the d -dimensional unit ball.

Following the standard k NN density-estimation principle of Loftsgaarden and Quesenberry (1965), we estimate the local density as:

$$\hat{p}(x) = \frac{K}{n V(x)}. \quad (9)$$

Observations lying in sparse or geometrically unusual regions will have low estimated density. We therefore define the nonconformity score as:

$$\alpha(x) = -\log \hat{p}(x), \quad (10)$$

so that less conforming observations receive larger scores.

Unlike standard Euclidean k NN density estimators, this construction adapts to the local covariance structure through ellipsoidal neighbourhoods, allowing the NCM to capture density shifts in multivariate streams, where Euclidean distances may provide a poor representation of local structure especially in high dimensions. The proposed estimator is based on the classical k -nearest neighbour density estimator of Loftsgaarden and Quesenberry (1965), with the usual local volume replaced by an ellipsoidal neighbourhood defined through Mahalanobis distance (Mahalanobis, 2018). To our knowledge, this ellipsoidal k NN density-based nonconformity measure has not previously been used for conformal martingale change detection.

5. Experiments and Results

This section experimentally evaluates the proposed feature-based Inductive Conformal Martingale approach for multivariate change-point detection. The aim is to assess its ability to detect distributional changes while maintaining low false alarm rates.

We evaluate the proposed method on four publicly available multivariate benchmark datasets: CASP, MiniBooNE, Sensorless Drive Diagnosis, and Credit Card. In each case, artificial abrupt changes are introduced into the monitoring stream in order to evaluate detection performance under controlled conditions.

We compare the proposed method with two nonparametric multivariate change-detection approaches: QuantTree (Boracchi et al., 2018) and QT-EWMA (Frittoli et al., 2023). These methods are particularly relevant because, like conformal martingales, they provide forms of false-alarm control, although the type of guarantee differs across methods.

5.1. Datasets

We evaluated the proposed method on four publicly available multivariate benchmark datasets, together with a real-world air-quality dataset used as a case study.

- **CASP (Protein Structure Prediction):** This dataset contains physicochemical properties of protein tertiary structures (Rana, 2013). Following preprocessing, the dataset consists of 45,730 instances described by 9 numerical features.
- **MiniBooNE (Particle Identification):** This dataset was created for particle identification in the MiniBooNE experiment (Roe, 2005). It consists of 130,064 instances described by 50 numerical features.
- **Sensorless Drive Diagnosis:** This dataset contains measurements from electric motor drives under different operating conditions (Bator, 2013). It consists of 58,509 instances described by 48 numerical features.
- **Credit Card Fraud:** This dataset contains anonymized credit card transaction features (Machine Learning Group, Université Libre de Bruxelles, 2013). It consists of 284,807 instances described by 29 numerical features.
- **Air Quality Dataset:** This dataset contains multivariate air-quality measurements collected over time, including gas concentrations, sensor responses, temperature, relative humidity, and absolute humidity (De Vito et al., 2008). We removed the NMHC_GT variable because it contained a large number of unavailable values encoded as -200 . For the remaining variables, all instances containing unavailable or missing values were removed. After preprocessing, the dataset consists of 6,941 instances described by 12 numerical features.

For the first four datasets, artificial abrupt changes were introduced into the monitoring stream in order to evaluate change-point detection performance under controlled conditions. The air-quality dataset was used as a real-world temporal case study.

5.2. Experimental Setting

For the first four benchmark datasets—CASP, MiniBooNE, Sensorless Drive Diagnosis, and Credit Card Fraud—we evaluated the proposed method under controlled artificial abrupt changes, following the general experimental framework of Boracchi et al. (2018). In each Monte Carlo repetition, the entire dataset was first randomly permuted and then partitioned into a reference set of size $TR = 4096$ and a disjoint monitoring pool. The reference set was used to construct the nonconformity measure. Two disjoint groups of size $\nu = 256$ were then selected from the monitoring pool to form the pre-change and unshifted post-change segments of the monitoring stream. The values of TR and ν are experimental choices and are not requirements of the proposed method.

All features were standardized using the mean and standard deviation estimated from the reference set. Let $\mathbf{Y}_i^{(0)} \in \mathbb{R}^d$, $i = 1, \dots, \nu$, denote the observations in the pre-change segment, and let $\mathbf{Y}_i^{(1)} \in \mathbb{R}^d$, $i = 1, \dots, \nu$, denote the unshifted observations selected for the post-change segment. An abrupt location shift was introduced by defining

$$\tilde{\mathbf{Y}}_i^{(1)} = \mathbf{Y}_i^{(1)} + \mathbf{\Delta}, \quad i = 1, \dots, \nu, \quad \mathbf{\Delta} \sim \mathcal{N}_d(\mathbf{0}, 0.1^2 I_d), \quad (11)$$

where I_d denotes the $d \times d$ identity matrix. A new d -dimensional shift vector $\mathbf{\Delta}$ was generated independently in each Monte Carlo repetition and applied to every observation in the corresponding post-change segment. Since the features were standardized before the change was introduced, the value 0.1 represents the standard deviation of the shift in standardized feature units. Thus, the direction and magnitude of the multivariate location shift varied across repetitions, while the shift remained fixed within each post-change segment.

These experiments focus on abrupt additive location changes in order to provide a controlled comparison with the experimental setting of [Boracchi et al. \(2018\)](#).

Except for the covariance-regularization sensitivity analysis, the proposed ellipsoidal k -nearest-neighbour density-based nonconformity measure was evaluated for $K = 1, \dots, 100$, with covariance regularization parameter $\lambda = 10^{-3}$. For the conformal martingale detector, we used the cautious betting function with parameter $\epsilon = 100$ and an alarm threshold of 100. By Ville’s inequality, this threshold provides an anytime false-alarm probability bound of 1% under exchangeability. The look-back window was set to $W = 5000$. All reported benchmark results are averages over 100 Monte Carlo repetitions.

We compared the proposed method with two competing nonparametric multivariate detectors: QuantTree and QT-EWMA. For QuantTree, we used 100 Monte Carlo repetitions, a reference set of size $TR = 4096$, and a test window of length $\nu = 256$. The number of QuantTree bins was set to 32, the significance level was set to $\alpha = 0.01$, and the total-variation statistic was used. No additional data transformation was applied. The same randomization and artificial location-shift procedure described above was used.

For QT-EWMA, we also used 100 Monte Carlo repetitions. In each repetition, the reference set had size $TR = 4096$, and the monitoring stream consisted of a pre-change segment of length 256, followed by a post-change segment of length 256. The same randomization and artificial location-shift procedure was used. The QT-EWMA parameters were set to $K = 32$, $\lambda = 0.1$, $ARL_0 = 1000$, and `min_tstar` = 5. The PCA transformation option was enabled. Detection thresholds were calibrated empirically under stationarity using 20,000 Monte Carlo repetitions.

For the air-quality dataset, no artificial change was introduced. Instead, the observations were retained in their original temporal order and used as a real-world case study for detecting possible violations of exchangeability over time. The first 3000 observations were used as the reference set, and the remaining observations were monitored sequentially. The ellipsoidal k -nearest-neighbour nonconformity measure was used with $K = 50$, covariance regularization parameter $\lambda = 10^{-3}$, and standardization based on the mean and standard deviation of the reference set. The cautious betting parameter was set to $\epsilon = 100$, and the alarm threshold was set to 100. The week variable was used only for visualization and interpretation of the detected changes.

For this real-world temporal case study, we compared the proposed method with QT-EWMA rather than QuantTree. QT-EWMA is designed for online sequential monitoring, whereas QuantTree is primarily a batch- or window-based change-detection method. Since the air-quality experiment aims to detect possible exchangeability violations as observations arrive over time, QT-EWMA provides the more appropriate comparison. The source code

and scripts used to reproduce the experiments reported in this paper are publicly available at <https://github.com/Pambos82/Multivariate-Change-Point-ICM>.

5.3. Results

This section evaluates the proposed feature-based conformal martingale detector on both controlled benchmark experiments and real-world sequential data. We first investigate sensitivity to the number of nearest neighbours K and the covariance regularization parameter λ , including a comparison between the proposed Ellipsoidal NCM and a conventional kNN-Euclidean NCM. We then compare the proposed approach against QuantTree and QT-EWMA in terms of detection rate, false-alarm rate, and detection delay. Finally, we present an air-quality case study illustrating the application of the method to real sequential sensor data.

5.3.1. SENSITIVITY TO THE NUMBER OF NEAREST NEIGHBOURS

To investigate the effect of the number of nearest neighbours, we repeated the experiments for $K = 1, \dots, 100$. For the proposed Ellipsoidal NCM, the covariance regularization parameter was fixed at $\lambda = 10^{-3}$. Figure 1 presents the sensitivity results with respect to K . The first column presents the detection rates, while the second column presents the corresponding false-alarm rates.

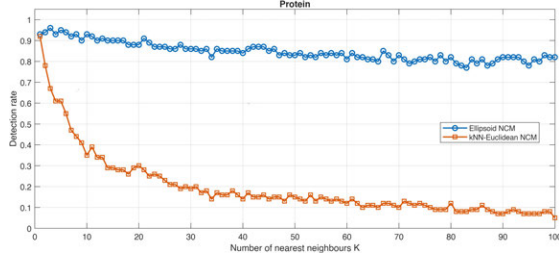
The proposed Ellipsoidal NCM is more robust to the choice of K than the kNN-Euclidean NCM. For the Credit and Sensorless datasets, the kNN-Euclidean NCM gives almost no detections for most values of K , whereas the Ellipsoidal NCM achieves high detection rates once K is sufficiently large. For the Protein dataset, the Ellipsoidal NCM maintains high detection performance over the whole range of K , while the kNN-Euclidean NCM deteriorates as K increases. On the Particle dataset, both methods achieve near-perfect detection, indicating that the induced change is comparatively easy to detect.

The false-alarm rates remain close to zero for both methods across all datasets and values of K . Therefore, the improvement of the Ellipsoidal NCM is not due to an increase in false alarms, but rather to improved sensitivity to distributional changes.

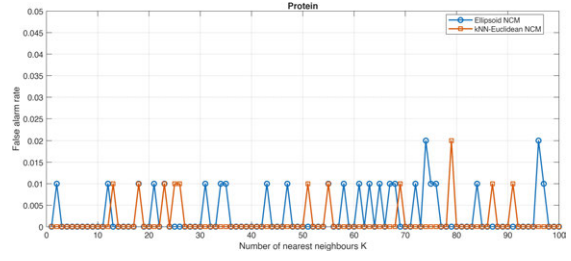
5.3.2. SENSITIVITY TO COVARIANCE REGULARIZATION

To investigate the sensitivity of the proposed Ellipsoidal NCM to the covariance regularization parameter, we repeated the experiments for $K = 1, \dots, 100$ and $\lambda \in \{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$. Figure 2 presents the detection and false-alarm rates for each combination of K and λ . The first column presents the detection rates, while the second column presents the corresponding false-alarm rates.

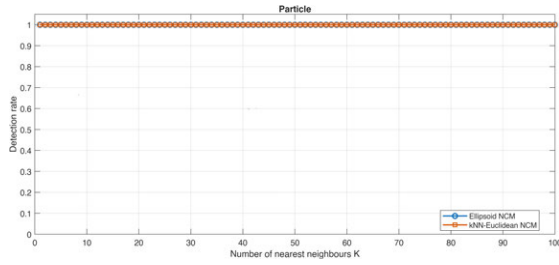
The results show that smaller regularization values generally provide the strongest detection performance. On the Protein dataset, $\lambda = 10^{-5}$ and $\lambda = 10^{-4}$ achieve the highest detection rates over most values of K , while performance gradually decreases as λ increases. On the Credit and Sensorless datasets, the values $\lambda \in \{10^{-5}, 10^{-4}, 10^{-3}\}$ achieve high detection rates once K is sufficiently large. In contrast, $\lambda = 10^{-2}$ produces substantially lower detection rates, while $\lambda = 10^{-1}$ results in almost no detections. On the Particle dataset, all considered values of λ achieve near-perfect detection, indicating that the introduced change is comparatively easy to detect.



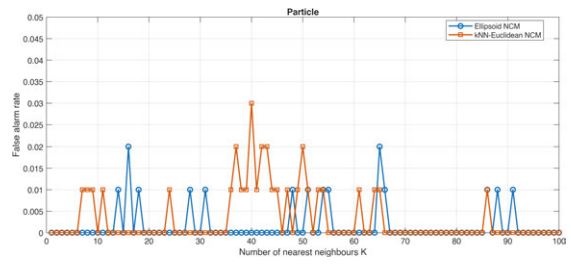
(a) Protein: detection rate



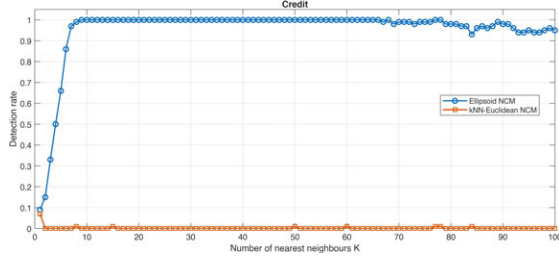
(b) Protein: false alarm rate



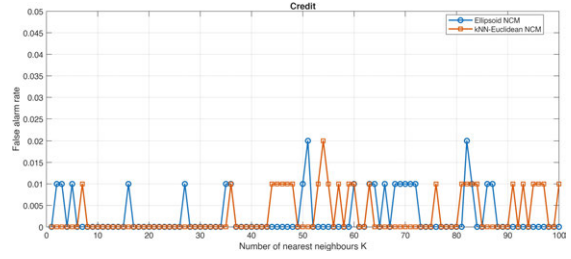
(c) Particle: detection rate



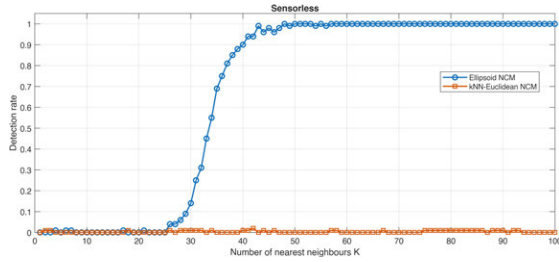
(d) Particle: false alarm rate



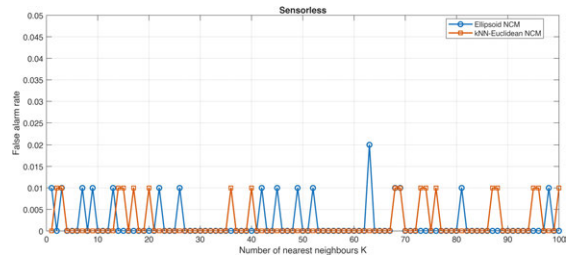
(e) Credit: detection rate



(f) Credit: false alarm rate



(g) Sensorless: detection rate



(h) Sensorless: false alarm rate

Figure 1: Detection and false alarm rates as a function of the number of nearest neighbours K . The proposed Ellipsoidal NCM is compared with the kNN-Euclidean NCM.

The deterioration observed for larger values of λ suggests that excessive regularization suppresses informative local covariance structure and causes the ellipsoidal neighbourhoods to become increasingly similar to spherical Euclidean neighbourhoods. Nevertheless, $\lambda = 10^{-3}$ provides good overall detection performance across the benchmark datasets while ensuring numerical stabilization of the local covariance matrices. We therefore retain $\lambda = 10^{-3}$ as the common setting used in the benchmark experiments.

The false-alarm rates remain close to zero across all datasets and parameter combinations, generally ranging between 0 and 0.02. Thus, the stronger detection performance obtained with smaller regularization values is not accompanied by a systematic increase in false alarms.

5.3.3. BENCHMARK RESULTS

Table 1 compares the proposed conformal martingale detector against QuantTree and QT-EWMA on the benchmark datasets. For the proposed method, we fixed $K = 50$ across all datasets rather than selecting a dataset-specific value. This value represents a common setting within the range for which the sensitivity analysis in Figure 1 shows consistently strong detection performance across the benchmark datasets, while the false alarm rates remain close to zero.

The proposed method achieved strong detection performance across the benchmark datasets, achieving perfect detection on MiniBooNE and near-perfect detection on the Sensorless and Credit datasets while maintaining very low false alarm rates between 0 and 0.01. These results are consistent with the validity guarantees provided by conformal martingales.

Compared with QuantTree, the proposed method achieved comparable or superior detection performance on the particle, sensorless, and credit datasets while maintaining lower or similar false alarm rates. QuantTree performed slightly better on the protein dataset.

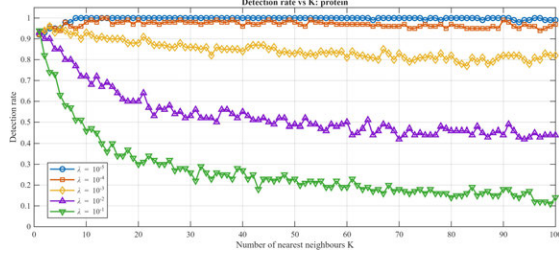
Compared with QT-EWMA, the proposed approach consistently produced lower false alarm rates while maintaining competitive or better detection performance, particularly on the credit dataset where QT-EWMA achieved much lower detection performance.

Regarding detection delay, the proposed conformal martingale method achieved fast detection on the Particle dataset (mean delay 15.1 samples) and moderate delays on the remaining datasets. While QT-EWMA often produced shorter delays, this came at the cost of significantly increased false alarm rates.

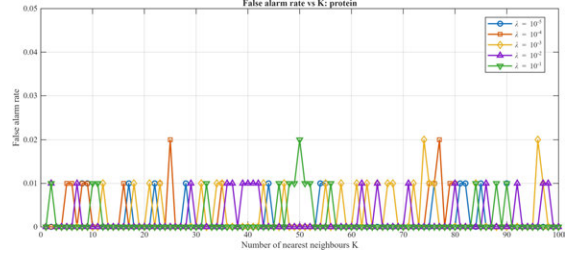
Overall, the results indicate that the proposed feature-based conformal martingale provides a favourable tradeoff between detection power, delay, and false alarm control.

5.3.4. AIR-QUALITY CASE STUDY

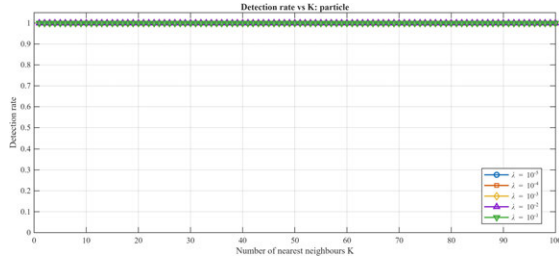
Figure 3 shows the evolution of the log-martingale on the air-quality dataset. A systematic increase is observed after approximately week 33, indicating sustained evidence against exchangeability. This suggests that the distribution of the sensor measurements changes after this period and is consistent with possible sensor degradation, recalibration effects, or other forms of temporal distribution shift. This observation is consistent with the analysis of De Vito et al. (2008), where a degradation in sensor-estimation performance was reported to begin around weeks 31–34.



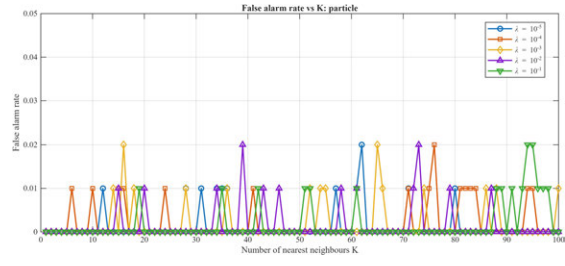
(a) Protein: detection rate



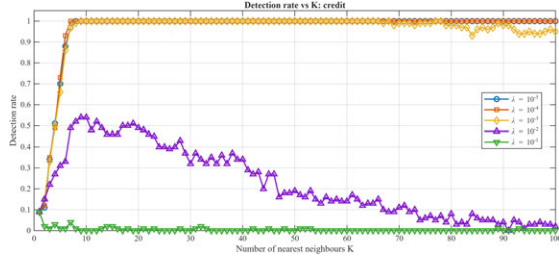
(b) Protein: false-alarm rate



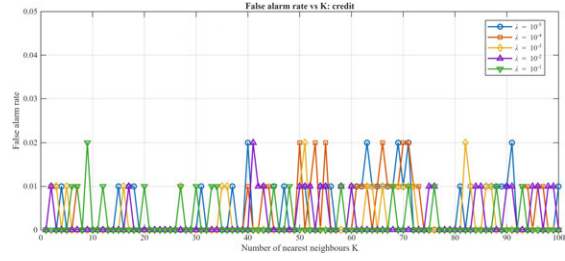
(c) Particle: detection rate



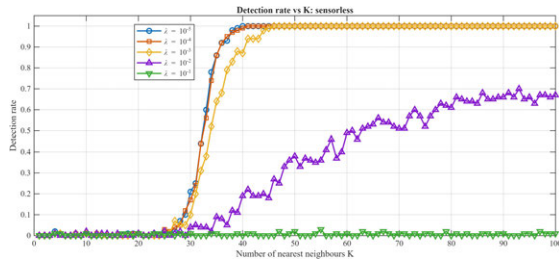
(d) Particle: false-alarm rate



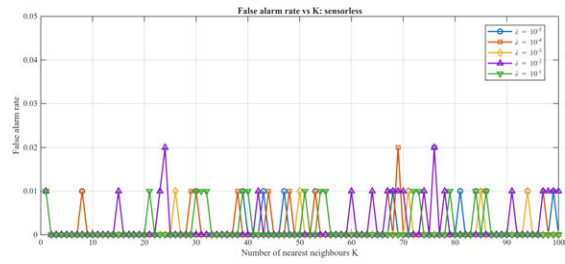
(e) Credit: detection rate



(f) Credit: false-alarm rate



(g) Sensorless: detection rate



(h) Sensorless: false-alarm rate

Figure 2: Detection and false-alarm rates of the proposed Ellipsoidal NCM as functions of the number of nearest neighbours K , for covariance regularization values $\lambda \in \{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$.

Table 1: Comparison with QuantTree and QT-EWMA

Dataset	Method	Detection	False Alarm	Mean Delay
Protein	Proposed	0.84	0.01	83.9
	QuantTree	0.86	0.00	–
	QT-EWMA	0.81	0.15	36.7
Particle	Proposed	1.00	0.01	15.1
	QuantTree	1.00	0.03	–
	QT-EWMA	0.91	0.09	3.2
Sensorless	Proposed	0.99	0.00	60.7
	QuantTree	0.78	0.01	–
	QT-EWMA	0.81	0.17	10.8
Credit	Proposed	0.99	0.01	83.5
	QuantTree	0.72	0.04	–
	QT-EWMA	0.46	0.17	81.6

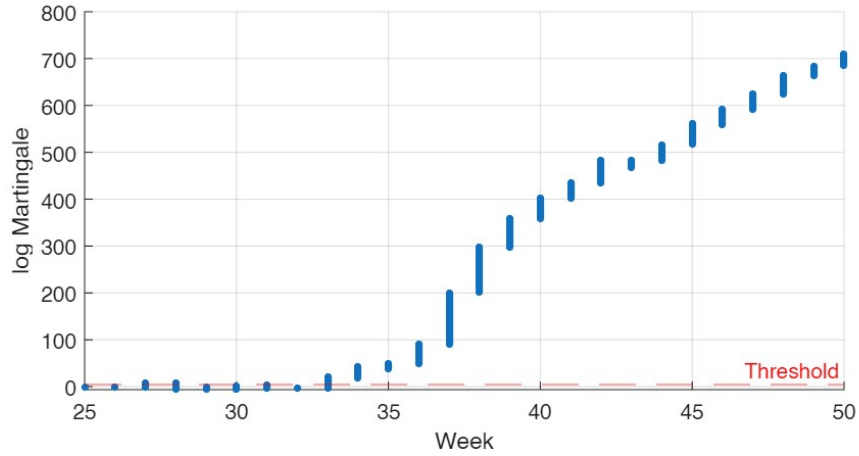


Figure 3: Log-martingale values over time for the air-quality dataset. The horizontal dashed line corresponds to a significance level of 1%.

A limitation of this case study is that temporally ordered air-quality measurements may exhibit serial dependence, seasonality, and gradual environmental variation. Such temporal structure may violate the exchangeability assumption even in the absence of a single abrupt change-point and may therefore affect the nominal false-alarm interpretation of the conformal martingale. Consequently, the detected increase should be interpreted as evidence of a departure from the reference exchangeable regime rather than as definitive evidence of a unique sensor recalibration point.

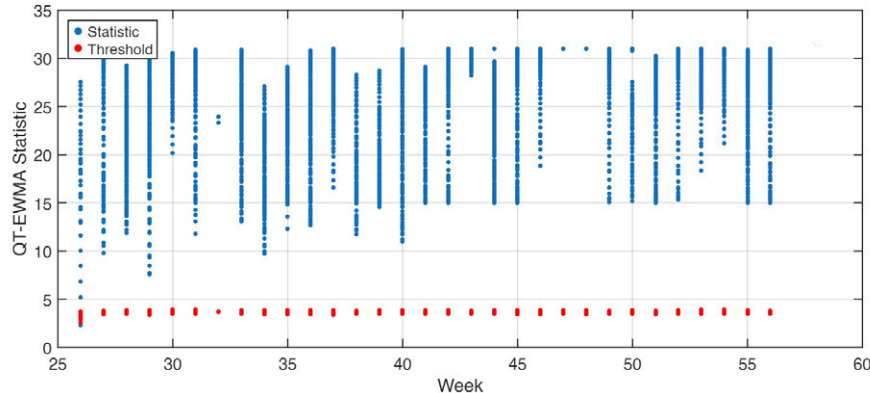


Figure 4: QT-EWMA monitoring statistic on the air-quality dataset. The dashed red line corresponds to the calibrated detection threshold.

Figure 4 shows the QT-EWMA monitoring statistic on the air-quality dataset. The statistic exceeds the calibrated threshold shortly after monitoring begins and remains above it for most of the subsequent period. This indicates persistent deviations from the reference distribution and suggests that the sensor measurements are no longer consistent with the initial calibration period.

However, QT-EWMA provides false-alarm control in terms of the average run length under the null rather than an anytime false-alarm probability bound. Its alarms should therefore be interpreted differently from those produced by conformal martingales. In particular, under the exchangeability assumption, the conformal martingale framework provides a bound on the probability of ever crossing the alarm threshold through Ville’s inequality, whereas QT-EWMA provides an ARL-based false-alarm control criterion.

6. Conclusions

In this work, we studied multivariate change-point detection using feature-based Inductive Conformal Martingales. We proposed a nonconformity measure based on an ellipsoidal k -nearest-neighbour density estimator that combines local density information with Mahalanobis distance and local covariance structure. This nonconformity measure allows conformal martingales to capture local geometric and density changes in multivariate data streams.

The experimental results on four benchmark datasets showed that the proposed method achieves high detection rates while maintaining low false alarm rates. The air-quality case

study further demonstrated the practical usefulness of the method, as the detected exchangeability violation was consistent with previously reported sensor degradation and recalibration needs.

Future work includes investigating additional feature representations and nonconformity measures, studying the theoretical properties of the proposed nonconformity measure, evaluating its performance under different shift magnitudes and covariance shifts, and extending the framework to gradual and recurring concept drift settings.

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